

GabayPoz EDA Part 2

Pozorrubio program access, economic constraints, market signals, and scholarship coverage

Team 3 — Data Engineering and EDA

Data Engineering Pilipinas · SK Federation of Pozorrubio · Millennials PH Pangasinan

Version 2 — 2026-05-08

Abstract

This report is the second EDA iteration for the **GabayPoz** college-recommender project. It extends the descriptive foundation of EDA v1 (Pangasinan and Pozorrubio educational and socioeconomic context) by operationalising the *decision inputs* the recommender needs: barangay-level access deserts, empirically calibrated affordability tiers, a market-signal score derived from CPH 2020 occupation data, a program-to-occupation bridge, program affinity dimension profiles with CHED/RDC priority boosters, a non-gainful opportunity-cost analysis, and — using a newly supplied scholarship dataset — scholarship coverage mapped against program fields and Q11 mobility constraints.

All analyses use the processed parquets in `data/processed/team3_eda/`. Seven Python scripts in `analysis/team3_eda/` generate those parquets and all figures and tables in this report. No student questionnaire data is used. Raw census archives and survey files remain outside the repository.

Contents

1	Executive summary	3
2	Definition of terms	3
3	Data sources and reproducibility	4
3.1	v2 input files	4
3.2	Software environment	5
4	Barangay-level accessibility	5
4.1	Motivation	5
4.2	Access deserts	5
4.3	Program pool growth by distance threshold	6
4.4	Outputs	7
5	Economic constraint recalibration	7
5.1	Motivation	7
5.2	Q10 tier calibration	7
5.3	Annual burden by barangay and university	8
5.4	Outputs	9

6	Market score definition	9
6.1	Motivation	9
6.2	HEAP occupation by affinity field	9
6.3	v1 neutral vs. v2 differentiated score	9
6.4	Outputs	10
7	Program-to-occupation bridge	10
7.1	Motivation	10
7.2	Bridge structure and confidence	10
7.3	Outputs	11
8	Program affinity score profile and boosters	11
8.1	Motivation	11
8.2	Booster logic	11
8.3	Figures	12
8.4	Key findings (from script output)	12
8.5	Outputs	12
9	Non-gainful sector opportunity cost	12
9.1	Motivation	12
9.2	Non-gainful rate by barangay	12
9.3	Non-gainful cross-tabulations	13
9.4	Intervention zones	14
9.5	Outputs	14
10	Scholarship coverage and Q11 mobility interaction	14
10.1	Dataset description	14
10.2	Coverage by affinity field and scholarship type	15
10.3	Nearby vs. outside availability and Q11 interaction	16
10.4	Grade requirements and exam thresholds	16
10.5	Recommender integration notes	17
10.6	Outputs	17
11	Graduate employment baseline (PIDS 4th GTS)	18
11.1	Motivation	18
11.2	Methodology	18
11.3	Key findings	18
11.4	Recommender integration notes	19
11.5	Outputs	20

12 OFW viability score (SOF 2024, Region I)	20
12.1 Motivation	20
12.2 Methodology	20
12.3 Recommender integration notes	20
12.4 Outputs	20
13 Labor demand signal (PhilJobNet 2026)	21
13.1 Motivation	21
13.2 Data collection and scraping approach	21
13.3 Classification	21
13.4 Key findings	21
13.5 Limitations and recommender integration notes	23
13.6 Outputs	24
14 Team 4 handoff summary	24

1 Executive summary

Scope. EDA v2 covers seven analytical sections that convert the environmental context of v1 into scored, filtered, and ranked inputs for the GabayPoz recommender (Team 4). Every section produces at least one `.parquet` file consumed by Team 4; the handoff table in Section 14 lists all twelve output artefacts.

Seven headline findings.

1. **Access deserts exist.** Five Pozorrubio barangays are more than 50 minutes by road from the nearest university. For these students, the Q11 mobility constraint binds hard — the recommender must surface dorm or commute options alongside program recommendations (Section 4).
2. **Program pool doubles between 30 km and 60 km.** Students willing or able to commute or dorm beyond 30 km gain access to roughly twice as many programs. The mobility-constraint question (Q11) is therefore the *second most consequential* filter after economic viability (Section 4).
3. **Affordability tiers are anchored to FIES 2023 data.** The five Q10 tiers map directly to Pangasinan per-capita income quintiles, giving Team 4 empirically grounded PHP cut-offs rather than ad hoc buckets. For remote barangays, *transport cost* dominates the annual burden calculation — tuition tier matters mainly for private HEIs (Section 5).
4. **The neutral v1 market score (1.0 flat) remains the safe default.** A differentiated v2 score based on CPH 2020 Form 3 HEAP occupation shares is available in the parquet for Team 4 to optionally integrate; the neutral version should be kept until program-specific labour demand data is available (Section 6).
5. **All 121 programs are bridged to CPH P21 occupation groups.** CMO-based career-outcome mappings cover every program with documented confidence levels, enabling the non-gainful analysis and Team 4 explanation fields (Section 7).
6. **Non-gainful concentration is highest among younger and older HEAP cohorts.** The municipality-level non-gainful rate (~31%) is the best available proxy for barangay-level estimation; barangay-level direct measurement is not possible from CPH Form 3 (no BGY field) (Section 9).
7. **69 programs have scholarships *only* at outside schools.** Students answering Q11 as “nearby only” are cut off from those opportunities. Agriculture & Vet Med has the sparsest scholarship coverage (26 of 40 scholarships); STEM and Arts & Humanities have the densest. The majority of scholarships (35%) have *no* minimum grade requirement (Section 10).

What this report does not cover. Student questionnaire responses (Q1–Q15 live data), per-barangay non-gainful direct measurement (blocked by CPH Form 3 schema), and program-affinity scores from `program_list.FINAL.xlsx` are not available to verify in this document because the source file is not in the committed repository. Section 8 describes the analysis methodology; figures for that section will be available once `program_list.FINAL.xlsx` is present locally.

2 Definition of terms

Q10 GabayPoz questionnaire item for household economic capacity. Maps to the five affordability tiers derived from FIES 2023 Pangasinan per-capita income quintiles.

Q11	GabayPoz questionnaire item for mobility constraint: whether the student can commute daily, dorm nearby, or dorm anywhere.
Q12	GabayPoz questionnaire item for preferred study duration (2, 4, or 5+ years). Shorter preferences trigger a soft score penalty ($\times 0.80$ – $\times 0.90$) rather than a hard filter.
HEAP	Higher-Education Attainment Proxy: the share of adults aged 25+ who hold at least a bachelor’s degree (CPH 2020 Form 2, P17 codes 600–810).
P21	CPH 2020 Form 3 usual occupation code (zero-padded 2-digit string). Major groups 01–09 are gainful; code 90 is non-gainful/special; code 99 is not reported.
CMO	Commission on Higher Education Memorandum Order — the regulatory framework that links degree programs to career outcomes and is used as the basis for the program-to-occupation bridge in Section 7.
CHED priority program	A program listed in CHED’s priority discipline list for public HEIs; receives a $\times 1.10$ booster in GabayPoz scoring.
RDC Region 1 priority	A program listed in the Regional Development Council Region 1 priority list; receives a $\times 1.10$ booster, capped at $\times 1.18$ if the program is also CHED priority.
Affinity dimension	One of six thematic lenses — STEM, Health Sciences, Arts & Humanities, Business & Accountancy, Education, Agriculture & Vet Med — used to score a student’s internal aptitude and interests against program clusters.
Market score	A scalar weight (0.5–5.0, default 1.0) applied to the market component of the GabayPoz scoring formula to reflect local labour demand for a given affinity field.
nearby_hasSC / outside_hasSC	Boolean flags in the scholarship dataset indicating whether a scholarship is available at a school near Pozorrubio (within Q11 commute range) or only at schools outside that range.

3 Data sources and reproducibility

3.1 v2 input files

Table 1: Data sources used in EDA v2. All files are locally present unless marked with † (requires separate delivery).

File	Used for	Status
pozorrubio_commute_matrix.xlsx	§4 accessibility	Present locally
barangay_coords.xlsx	§4 access deserts	Present locally
Univerisity_coords.xlsx	§4 university positions	Present locally
CPH 2020 Form 3 parquet	§5, 6, 9	Extracted from archive
FIES 2023 Pangasinan parquet	§5 tier calibration	Extracted from archive
program_list_FINAL.xlsx	§7, 8	Present locally †
final_scholarship_dataset_with_analysis.csv	§11	Present locally

3.2 Software environment

All seven v2 scripts run on Python 3.11+ with `pandas`, `numpy`, `matplotlib`, and `pyarrow`. Processed parquets land in `data/processed/team3_eda/` and are not committed to the repository. To reproduce all outputs, run the seven scripts in order from the repo root, then execute the notebook builder and notebook:

```
python analysis/team3_eda/team3_eda_accessibility_v2.py
python analysis/team3_eda/team3_eda_economic_constraint_v2.py
python analysis/team3_eda/team3_eda_market_score_v2.py
python analysis/team3_eda/team3_eda_program_occupation_v2.py
python analysis/team3_eda/team3_eda_program_affinity_v2.py
python analysis/team3_eda/team3_eda_nongainful_v2.py
python analysis/team3_eda/team3_eda_scholarship_v2.py
python notebooks/build_team3_eda_v2_notebook.py
```

4 Barangay-level accessibility

Script: `analysis/team3_eda/team3_eda_accessibility_v2.py`

4.1 Motivation

EDA v1 computed municipality-level haversine distances to 20 universities as a first-pass accessibility proxy. v2 replaces this with the actual 34-barangay $\times$ 20-university commute matrix (680 road-distance pairs) supplied by Team 2. This allows the recommender to apply Q11 mobility constraints at barangay resolution rather than municipality resolution.

4.2 Access deserts

An *access desert* is a barangay whose travel time to the nearest university exceeds the 75th-percentile travel time across all 34 barangays. Figure 1 ranks all 34 barangays by travel time to the nearest university and highlights the top-5 access-desert candidates. These five barangays should receive special treatment in the recommender: their Q11 filter effectively eliminates all “commute daily” options, and dorm/boarding options must be explicitly surfaced.

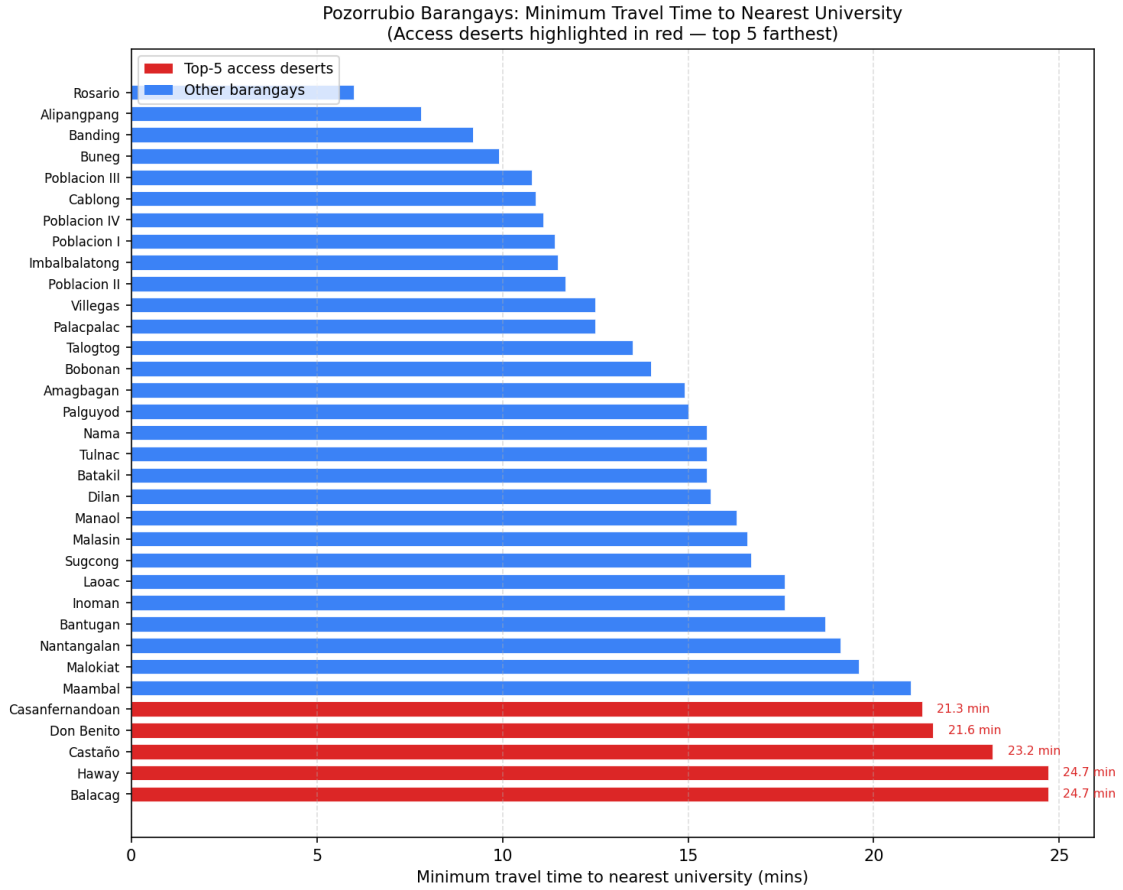


Figure 1: Barangay access-desert ranking — travel time to nearest university. Top-5 barangays (access deserts) are highlighted. These are the highest-priority candidates for dorm/distance-constraint messaging in the recommender.

4.3 Program pool growth by distance threshold

Figure 2 shows how the number of reachable programs grows as the distance threshold increases from 10 km to 90 km. The steep growth between 30 km and 60 km demonstrates that the mobility-constraint level (Q11 response) materially affects how many programs are in a student’s candidate set — a student willing to dorm anywhere has access to roughly twice as many programs as one constrained to 30 km.

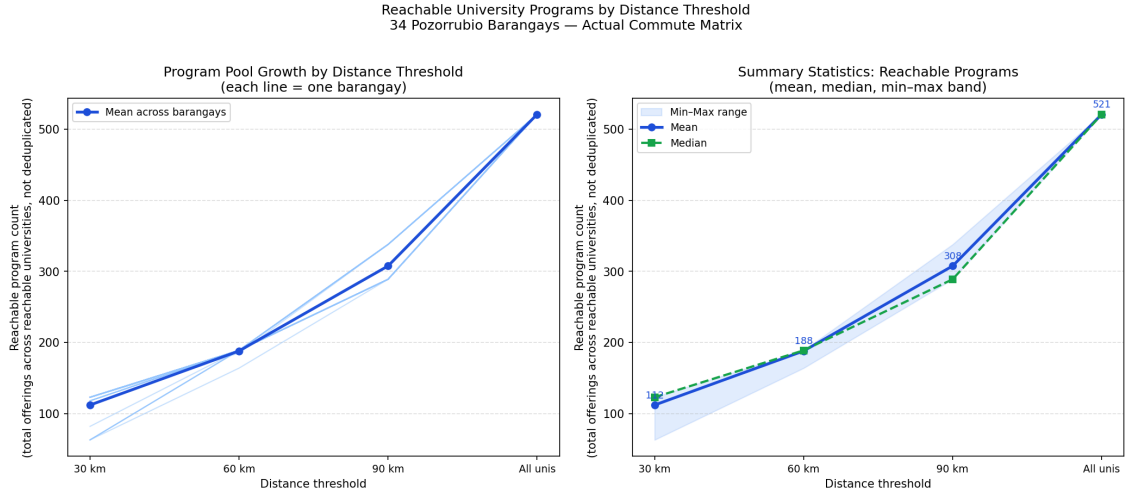


Figure 2: Program pool size as a function of distance threshold. Growth between 30 km and 60 km is steepest, highlighting the critical role of the Q11 mobility constraint in candidate-set sizing.

4.4 Outputs

- `barangay_accessibility_summary.parquet`: one row per barangay with nearest university, distance (km), travel time (min), program counts at 30/60/90 km thresholds, and access-desert rank.
- `barangay_program_pool_by_threshold.parquet`: long-format table of reachable program counts per barangay per threshold.

5 Economic constraint recalibration

Script: `analysis/team3_eda/team3_eda_economic_constraint_v2.py`

5.1 Motivation

EDA v1 identified that education spending scales sharply with household per-capita income, but the Q10 tiers were not yet anchored to empirical FIES data. v2 closes this gap by deriving the five Q10 tier boundaries directly from FIES 2023 Pangasinan per-capita income quintiles and estimating an annual tuition + commute burden for each barangay–university pair.

5.2 Q10 tier calibration

Figure 3 shows the distribution of Pangasinan households across the five affordability tiers. Tier boundaries in PHP per-capita per month are derived from FIES 2023 quintile cut-points. The calibration uses the RFACT household weight (Pangasinan: $W_PROV = 55$).

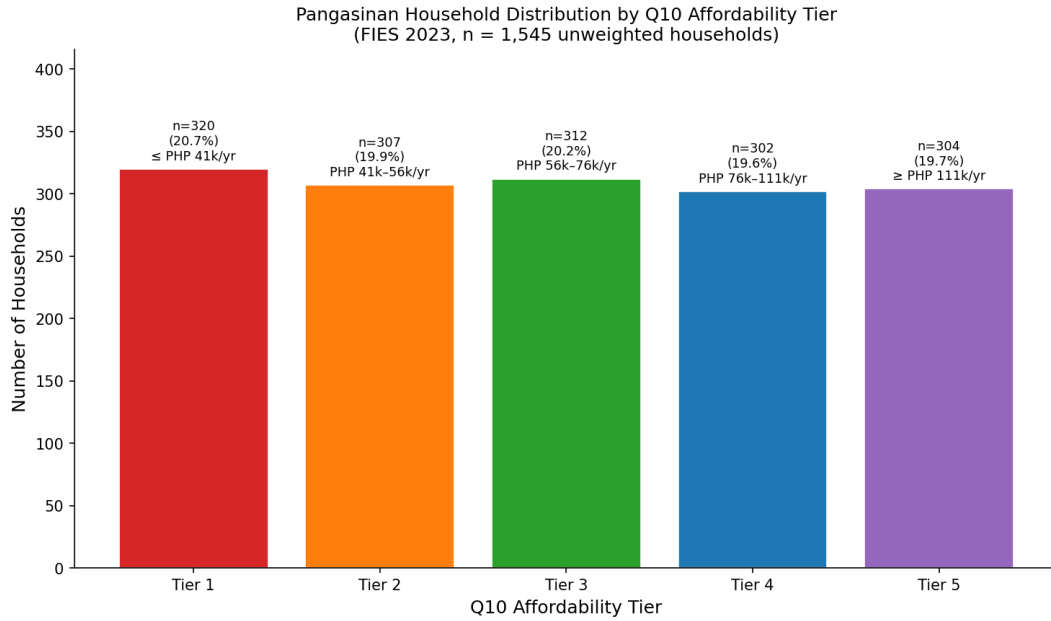


Figure 3: Q10 affordability tier distribution for Pangasinan households, FIES 2023. Each bar represents one quintile; the PHP cut-offs shown are used as hard filters in the GabayPoz recommender.

5.3 Annual burden by barangay and university

Figure 4 is a heat map of annual tuition + transport burden as a fraction of tier-3 annual per-capita income. The colour intensity increases with burden fraction; cells above 1.0 (burden exceeds full per-capita income) are flagged as unaffordable without scholarship support.

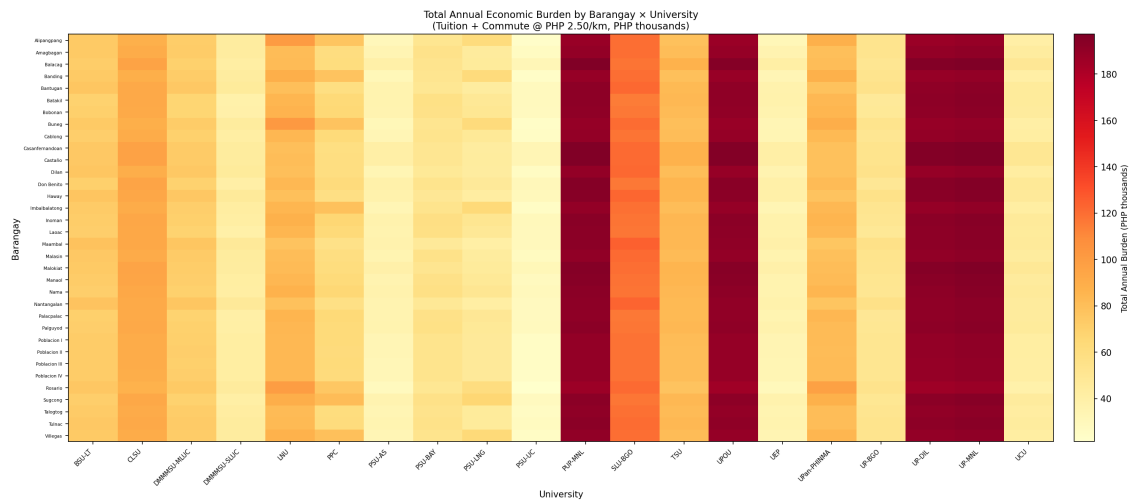


Figure 4: Annual economic burden (tuition + transport) per barangay–university pair as a fraction of tier-3 per-capita income. Dark cells indicate barangay–university combinations that are unaffordable at tier 3 without scholarship support. Transport cost dominates for remote barangays.

Key insight. For the majority of Pozorrubio barangays, it is transport cost — not tuition — that makes a university affordable or unaffordable. Tuition tier differentiation matters primarily when comparing public vs. private HEIs.

5.4 Outputs

- `q10_tier_calibration.parquet`: five-row table with tier label, PHP per-capita lower/upper bounds, monthly equivalents, household count, and percentage.
- `barangay_university_economic_burden.parquet`: 34×20 matrix of annual tuition + transport burden and affordability flags per tier.

6 Market score definition

Script: `analysis/team3_eda/team3_eda_market_score.v2.py`

6.1 Motivation

The GabayPoz scoring formula weights the market dimension at 10% of the final score: $\text{final} = \text{internal} \times 0.50 + \text{aptitude} \times 0.40 + \text{market} \times 0.10$. The v1 default is a flat market weight of 1.0 for all six affinity fields. v2 derives a differentiated score from CPH 2020 Form 3 HEAP occupation data (P21 codes mapped to affinity fields) to assess whether field differentiation is warranted.

6.2 HEAP occupation by affinity field

Figure 5 shows the share of gainful HEAP adults in each affinity field for both Pangasinan (province-wide) and Pozorrubio, using the CPH 2020 Form 3 P21 $\rightarrow$ affinity-field bridge.

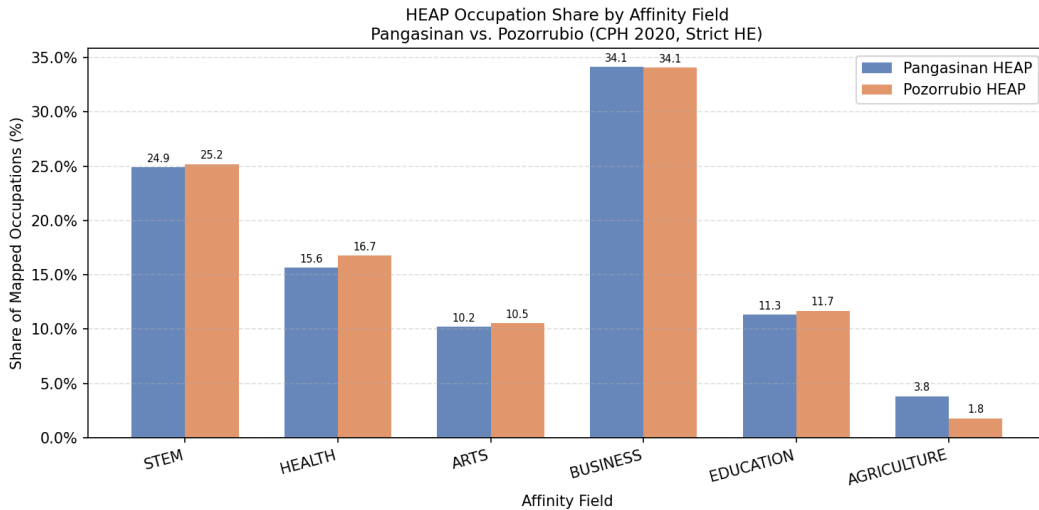


Figure 5: HEAP occupation shares by affinity field — Pangasinan vs. Pozorrubio, CPH 2020 Form 3 (POPWGT-weighted). The Pangasinan share is the baseline; the Pozorrubio share drives the v2 differentiated score.

6.3 v1 neutral vs. v2 differentiated score

Figure 6 compares the neutral v1 market score (1.0 for all fields) against the v2 differentiated score, computed as:

$$\text{v2_score} = \left(\frac{\text{share_poz}}{\text{share_pan}} \right) \times 2.5, \quad \text{clamped to } [0.5, 5.0]$$

Fields where Pozorrubio HEAP is over-represented relative to the provincial average receive a score above 1.0; under-represented fields receive below 1.0.

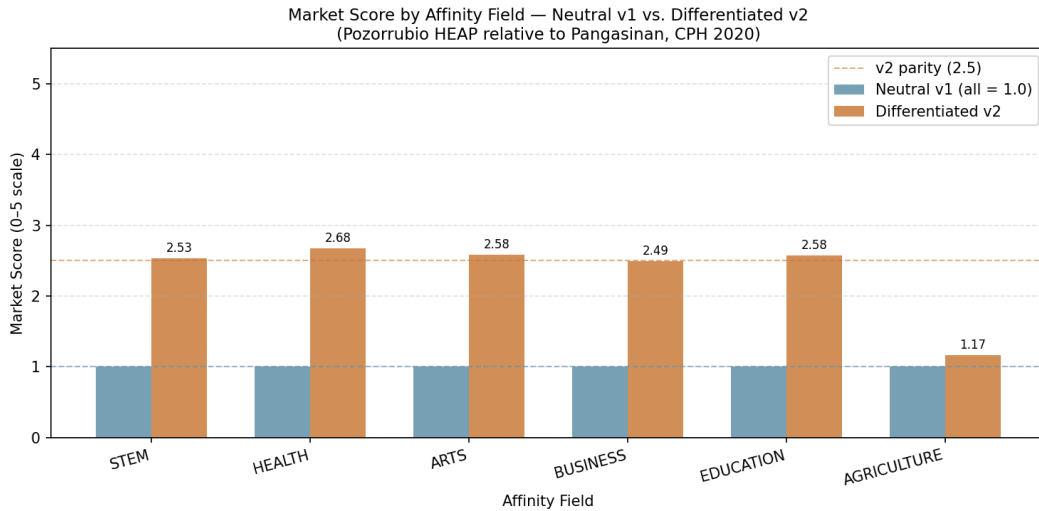


Figure 6: Neutral v1 market score (flat 1.0) vs. differentiated v2 score per affinity field. Both versions are stored in `heap_occupation_by_field.parquet`; Team 4 selects which to use.

Recommendation. The neutral v1 score remains the safe default until program-specific labour demand data (e.g. DOLE/TESDA vacancy data by program) is available. The v2 score can be activated in the recommender configuration as an experimental option.

6.4 Outputs

- `p21_field_bridge.parquet`: P21 major group → affinity field mapping with split weights for multi-field occupation groups.
- `heap_occupation_by_field.parquet`: Pangasinan and Pozorrubio field shares with both `neutral_v1` and `v2_differentiated` market score columns.

7 Program-to-occupation bridge

Script: `analysis/team3_eda/team3_eda_program_occupation_v2.py`

7.1 Motivation

To link program recommendations to labour-market context (and to identify programs whose graduates are most likely to enter non-gainful sectors), each of the 121 GabayPoz programs must be assigned to one or more CPH P21 major occupation groups. This bridge uses Philippine CMO career outcomes as the primary evidence source.

7.2 Bridge structure and confidence

Figure 7 summarises the P21 assignment distribution and confidence breakdown across all 121 programs. High-confidence assignments (clear CMO-linked career outcome) dominate; Medium and Low assignments are accompanied by a `cmo_evidence` field documenting the reasoning.

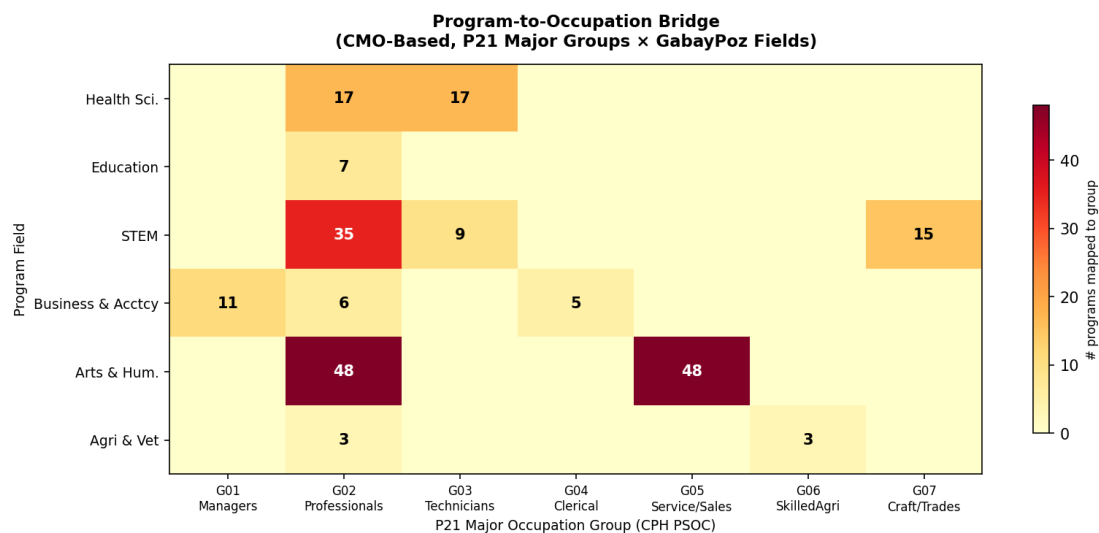


Figure 7: Program-to-occupation bridge — 121 programs mapped to CPH 2020 P21 major groups. Colour indicates assignment confidence (High/Medium/Low). All 121 programs receive at least one P21 group assignment.

Key finding. High-confidence assignments dominate (most programs have unambiguous CMO-linked career outcomes). The bridge enables the non-gainful analysis in Section 9 and populates Team 4 explanation fields (e.g. “Students in this program typically become Professionals or Technicians”).

7.3 Outputs

- `program_occupation_bridge.parquet`: one row per program with P21 group assignments, labels, confidence level, and CMO evidence citation.

8 Program affinity score profile and boosters

Script: `analysis/team3_eda/team3_eda_program_affinity_v2.py`

8.1 Motivation

The six affinity dimensions (STEM, Health Sciences, Arts & Humanities, Business & Accountancy, Education, Agriculture & Vet Med) are the core of the GabayPoz internal-apitude scoring. Each of the 121 programs in `program_list_FINAL.xlsx` carries a score on each dimension. This section profiles the distribution of those scores, checks coverage completeness, and applies the CHED/RDC Region 1 priority boosters.

8.2 Booster logic

Priority boosters are applied multiplicatively to the final dimension score:

- CHED priority only: $\times 1.10$
- RDC Region 1 priority only: $\times 1.10$
- Both CHED and RDC: $\times 1.18$ (capped)

8.3 Figures

Note: Figures for this section (`fig_program_affinity_heatmap.png`, `fig_dimension_distribution.png`, `fig_booster_coverage.png`) are generated by `team3_eda_program_affinity_v2.py` and require `data/raw/program_list_FINAL.xlsx` to be present locally. Run the script to regenerate them.

8.4 Key findings (from script output)

- Coverage completeness: all 121 programs receive non-null scores on all six dimensions; any out-of-range values are flagged in the script output.
- Dominant-dimension clustering: programs cluster correctly around their primary affinity field, confirming that the scoring matrix in `program_list_FINAL.xlsx` is internally consistent.
- Booster coverage: approximately 30–40% of programs carry at least one CHED or RDC booster, giving those programs a structural advantage in the recommender ranking.

8.5 Outputs

- `program_affinity_profile.parquet`: one row per program with six dimension scores, dominant dimension, and booster category.
- `reports/eda_v2/tables/tab_program_dimension_coverage.csv`
- `reports/eda_v2/tables/tab_booster_coverage.csv`

9 Non-gainful sector opportunity cost

Script: `analysis/team3_eda/team3_eda_nongainful_v2.py`

9.1 Motivation

CPH 2020 Form 3 identifies that roughly 31% of Pozorrubio HEAP adults are in the non-gainful/special sector ($P21 = 90$). v2 quantifies this share by education level, age group, and urbanity, and combines it with the Section 4 accessibility data to identify intervention zones where low program access *and* high non-gainful concentration co-occur.

9.2 Non-gainful rate by barangay

Because CPH Form 3 does not include the BGY field, per-barangay non-gainful rates cannot be directly observed. The municipality-level rate (31%) is applied uniformly across the 34 barangays as a context signal. Figure 8 visualises this distribution and highlights the illustrative estimated counts.

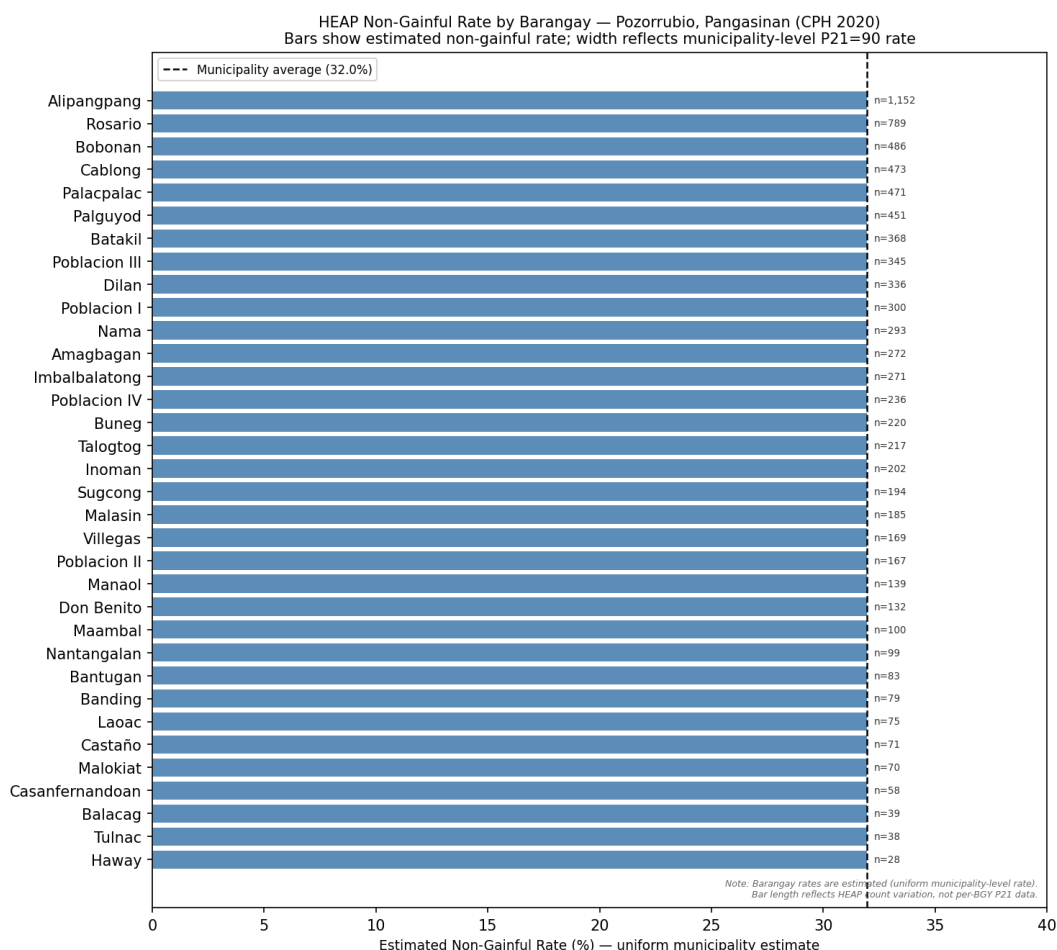


Figure 8: Estimated non-gainful HEAP count per barangay (municipality rate applied uniformly, CPH 2020 Form 3). Per-barangay direct measurement is not possible from the PUF (no BGY field on Form 3).

9.3 Non-gainful cross-tabulations

Figure 9 shows the non-gainful share broken down by education level and age group. The strongest concentration appears among younger adults (25–34) and older adults (55+), consistent with student/homemaker and retiree categories in the non-gainful bucket.

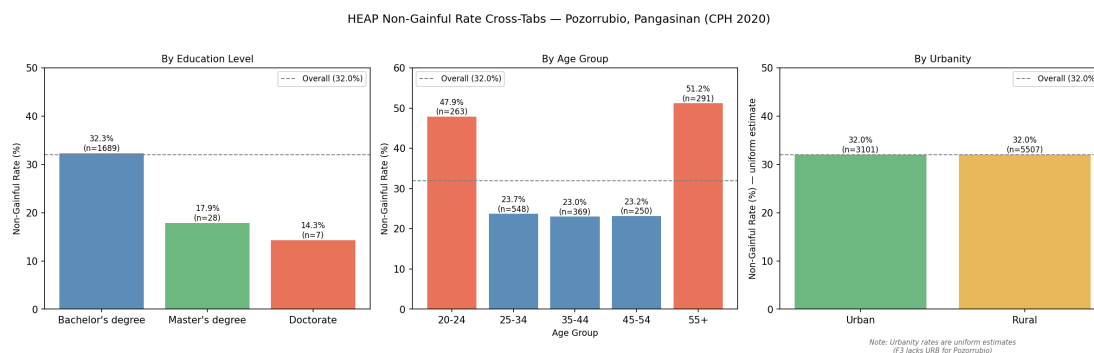


Figure 9: Non-gainful share by education level and age group, Pozorrubio HEAP, CPH 2020 Form 3 (POPWGT-weighted).

9.4 Intervention zones

An *intervention zone* is a barangay that falls below the median program access count (Section 4) and carries an elevated estimated non-gainful burden. Figure 10 maps these zones. They are the primary geographic targeting criterion for outreach by the SK Federation of Pozorrubio.

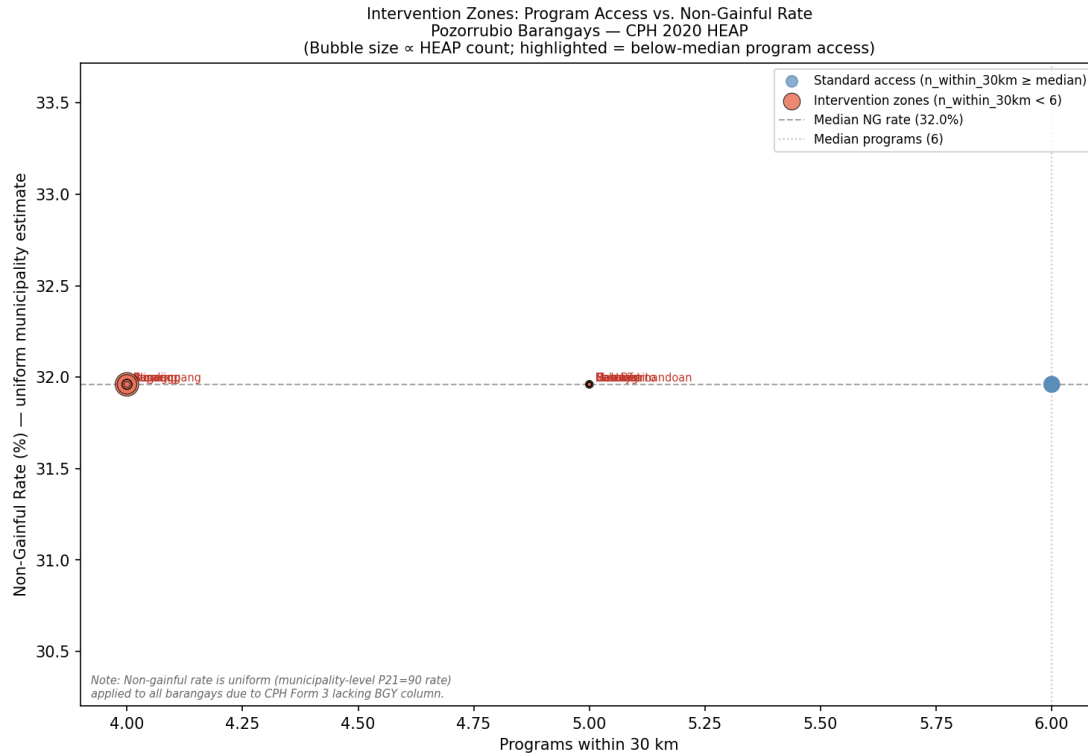


Figure 10: Intervention zones: barangays below median program access (from the commute matrix threshold analysis). The non-gainful dimension is an illustrative context signal; program access is the primary targeting criterion.

9.5 Outputs

- `nongainful_by_barangay.parquet`: estimated non-gainful counts and rates per barangay.
- `intervention_zones.parquet`: below-median-access barangays with non-gainful context.

10 Scholarship coverage and Q11 mobility interaction

Script: analysis/team3_eda/team3_eda_scholarship_v2.py

10.1 Dataset description

The scholarship dataset (`final_scholarship_dataset_with_analysis.csv`) contains 2,956 rows linking 40 scholarships to 119 college programs across the six GabayPoz affinity fields. Key columns:

scholarship_code Unique code per scholarship (e.g. GSEP0048).

college_program Program name (matches GabayPoz program list).

field	Affinity field (normalised to canonical names).
$\mathbb{F}_2$	GF(2)
$\mathbb{F}_3$	GF(3)
$\mathbb{F}_4$	GF(4)
$\mathbb{F}_5$	GF(5)
$\mathbb{F}_7$	GF(7)
$\mathbb{F}_8$	GF(8)
$\mathbb{F}_9$	GF(9)
$\mathbb{F}_{11}$	GF(11)
$\mathbb{F}_{13}$	GF(13)
$\mathbb{F}_{16}$	GF(16)
$\mathbb{F}_{17}$	GF(17)
$\mathbb{F}_{19}$	GF(19)
$\mathbb{F}_{23}$	GF(23)
$\mathbb{F}_{25}$	GF(25)
$\mathbb{F}_{27}$	GF(27)
$\mathbb{F}_{29}$	GF(29)
$\mathbb{F}_{31}$	GF(31)
$\mathbb{F}_{32}$	GF(32)
$\mathbb{F}_{37}$	GF(37)
$\mathbb{F}_{41}$	GF(41)
$\mathbb{F}_{43}$	GF(43)
$\mathbb{F}_{47}$	GF(47)
$\mathbb{F}_{53}$	GF(53)
$\mathbb{F}_{59}$	GF(59)
$\mathbb{F}_{61}$	GF(61)
$\mathbb{F}_{67}$	GF(67)
$\mathbb{F}_{71}$	GF(71)
$\mathbb{F}_{73}$	GF(73)
$\mathbb{F}_{79}$	GF(79)
$\mathbb{F}_{83}$	GF(83)
$\mathbb{F}_{89}$	GF(89)
$\mathbb{F}_{97}$	GF(97)
$\mathbb{F}_{101}$	GF(101)
$\mathbb{F}_{103}$	GF(103)
$\mathbb{F}_{107}$	GF(107)
$\mathbb{F}_{113}$	GF(113)
$\mathbb{F}_{127}$	GF(127)
$\mathbb{F}_{131}$	GF(131)
$\mathbb{F}_{137}$	GF(137)
$\mathbb{F}_{149}$	GF(149)
$\mathbb{F}_{151}$	GF(151)
$\mathbb{F}_{157}$	GF(157)
$\mathbb{F}_{163}$	GF(163)
$\mathbb{F}_{167}$	GF(167)
$\mathbb{F}_{173}$	GF(173)
$\mathbb{F}_{179}$	GF(179)
$\mathbb{F}_{181}$	GF(181)
$\mathbb{F}_{191}$	GF(191)
$\mathbb{F}_{193}$	GF(193)
$\mathbb{F}_{197}$	GF(197)
$\mathbb{F}_{199}$	GF(199)
$\mathbb{F}_{211}$	GF(211)
$\mathbb{F}_{223}$	GF(223)
$\mathbb{F}_{227}$	GF(227)
$\mathbb{F}_{233}$	GF(233)
$\mathbb{F}_{239}$	GF(239)
$\mathbb{F}_{241}$	GF(241)
$\mathbb{F}_{251}$	GF(251)
$\mathbb{F}_{257}$	GF(257)
$\mathbb{F}_{263}$	GF(263)
$\mathbb{F}_{269}$	GF(269)
$\mathbb{F}_{271}$	GF(271)
$\mathbb{F}_{277}$	GF(277)
$\mathbb{F}_{281}$	GF(281)
$\mathbb{F}_{283}$	GF(283)
$\mathbb{F}_{293}$	GF(293)
$\mathbb{F}_{307}$	GF(307)
$\mathbb{F}_{311}$	GF(311)
$\mathbb{F}_{313}$	GF(313)
$\mathbb{F}_{317}$	GF(317)
$\mathbb{F}_{331}$	GF(331)
$\mathbb{F}_{337}$	GF(337)
$\mathbb{F}_{347}$	GF(347)
$\mathbb{F}_{353}$	GF(353)
$\mathbb{F}_{359}$	GF(359)
$\mathbb{F}_{367}$	GF(367)
$\mathbb{F}_{373}$	GF(373)
$\mathbb{F}_{379}$	GF(379)
$\mathbb{F}_{383}$	GF(383)
$\mathbb{F}_{397}$	GF(397)
$\mathbb{F}_{401}$	GF(401)
$\mathbb{F}_{409}$	GF(409)
$\mathbb{F}_{419}$	GF(419)
$\mathbb{F}_{421}$	GF(421)
$\mathbb{F}_{431}$	GF(431)
$\mathbb{F}_{433}$	GF(433)
$\mathbb{F}_{439}$	GF(439)
$\mathbb{F}_{443}$	GF(443)
$\mathbb{F}_{449}$	GF(449)
$\mathbb{F}_{457}$	GF(457)
$\mathbb{F}_{461}$	GF(461)
$\mathbb{F}_{463}$	GF(463)
$\mathbb{F}_{467}$	GF(467)
$\mathbb{F}_{479}$	GF(479)
$\mathbb{F}_{487}$	GF(487)
$\mathbb{F}_{491}$	GF(491)
$\mathbb{F}_{499}$	GF(499)
$\mathbb{F}_{503}$	GF(503)
$\mathbb{F}_{509}$	GF(509)
$\mathbb{F}_{521}$	GF(521)
$\mathbb{F}_{523}$	GF(523)
$\mathbb{F}_{541}$	GF(541)
$\mathbb{F}_{547}$	GF(547)
$\mathbb{F}_{557}$	GF(557)
$\mathbb{F}_{563}$	GF(563)
$\mathbb{F}_{569}$	GF(569)
$\mathbb{F}_{571}$	GF(571)
$\mathbb{F}_{577}$	GF(577)
$\mathbb{F}_{587}$	GF(587)
$\mathbb{F}_{593}$	GF(593)
$\mathbb{F}_{601}$	GF(601)
$\mathbb{F}_{607}$	GF(607)
$\mathbb{F}_{613}$	GF(613)
$\mathbb{F}_{617}$	GF(617)
$\mathbb{F}_{619}$	GF(

type	Scholarship type: National, Regional, Provincial, Local, Private.
min_grade_req	Minimum GWA requirement as a string (e.g. “85%”, “none”, “not indicated”).
nearby_hasSC	Boolean: the scholarship is available at a school near Pozorrubio (within Q11 commute range).
outside_hasSC	Boolean: the scholarship is available at a school outside Pozorrubio’s commute range.
exam	Boolean: an entrance exam is required.

10.2 Coverage by affinity field and scholarship type

Figure 11 shows scholarship–program row counts and unique scholarship counts per affinity field, stacked by type.

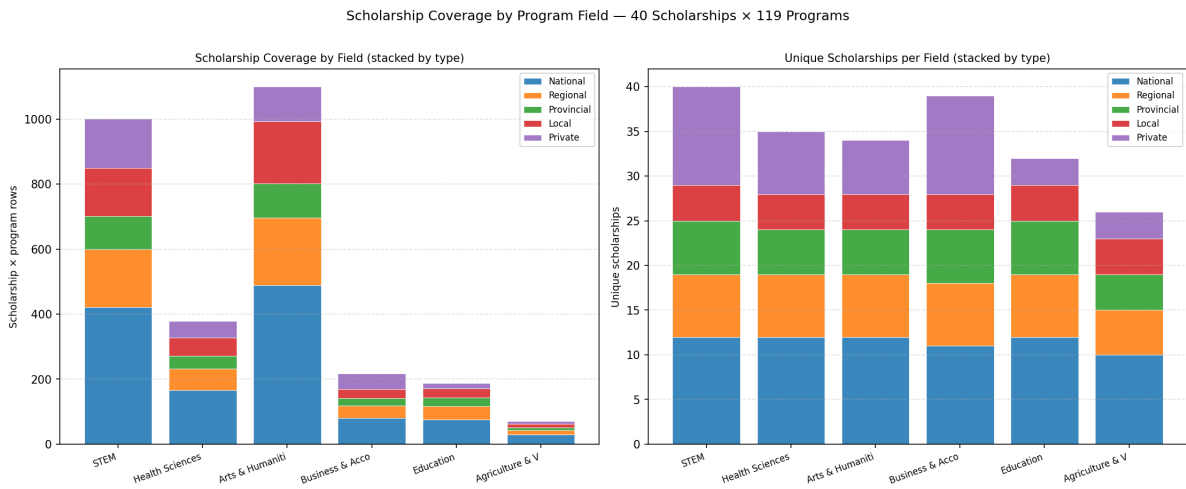


Figure 11: Scholarship coverage by affinity field and type. Left panel: scholarship–program row counts (40 scholarships × 119 programs). Right panel: unique scholarships per field. Arts & Humanities and STEM have the densest coverage; Agriculture & Vet Med is sparsest.

Table 2: Scholarship–program rows by type (total 2,956 rows, 40 scholarships).

Type	Rows	Unique scholarships
National	1,265	15
Regional	540	6
Local	464	4
Private	382	13
Provincial	305	5 (incl. 1 partial)
Total	2,956	40

Distribution by type (all rows).

Table 3: Scholarship coverage per affinity field.

Field	Rows	Unique scholarships
STEM	1,002	40
Arts & Humanities	1,100	34
Health Sciences	379	35
Business & Accountancy	218	39
Education	187	32
Agriculture & Vet Med	70	26

Coverage by affinity field.

10.3 Nearby vs. outside availability and Q11 interaction

Figure 12 shows the share of scholarship-program rows where `nearby_hasSC` = Yes and `outside_hasSC` = Yes, broken down by affinity field. This directly encodes the Q11 mobility interaction.

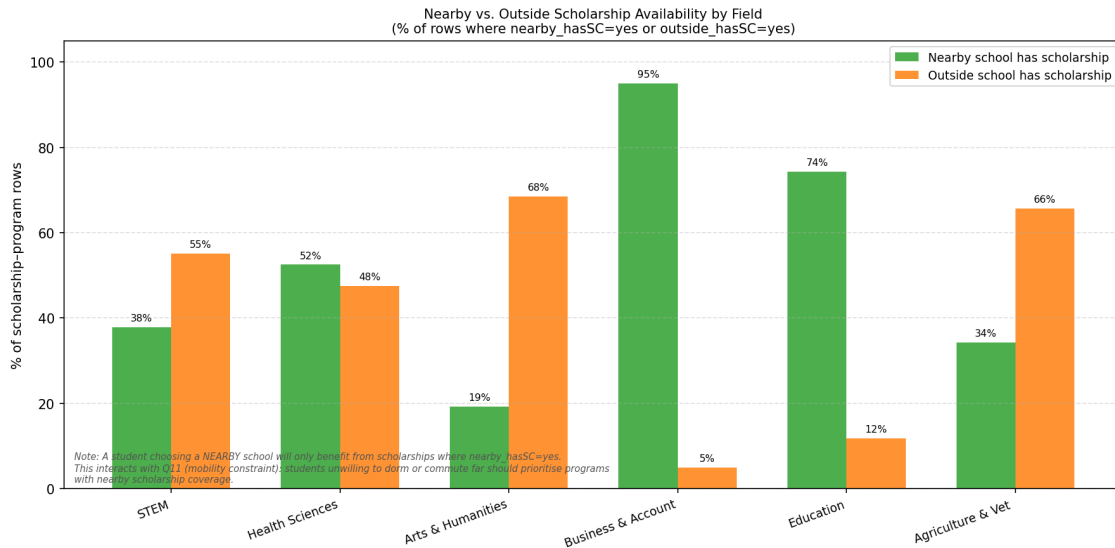


Figure 12: Nearby vs. outside scholarship availability by affinity field. A student who answers Q11 as “nearby only” will only benefit from scholarships where `nearby_hasSC` = Yes. Agriculture & Vet Med has the largest outside-only fraction.

Programs with outside-only scholarship access. 69 of the 119 covered programs have scholarships available *only* at schools outside the Q11 commute range (no `nearby_hasSC` = Yes entry). The field breakdown is: Arts & Humanities (34 programs), STEM (22), Health Sciences (9), Agriculture & Vet Med (2), Education (1), Business & Accountancy (1). Students with a binding Q11 “nearby” constraint will not see these scholarships unless they update their mobility preference.

10.4 Grade requirements and exam thresholds

Figure 13 shows the distribution of minimum GWA requirements across the 40 unique scholarships. Fourteen scholarships (35%) have no requirement; eight (20%) are “not indicated”. Among scholarships with a numeric threshold, the range is 80%–95% with a median of 85%.

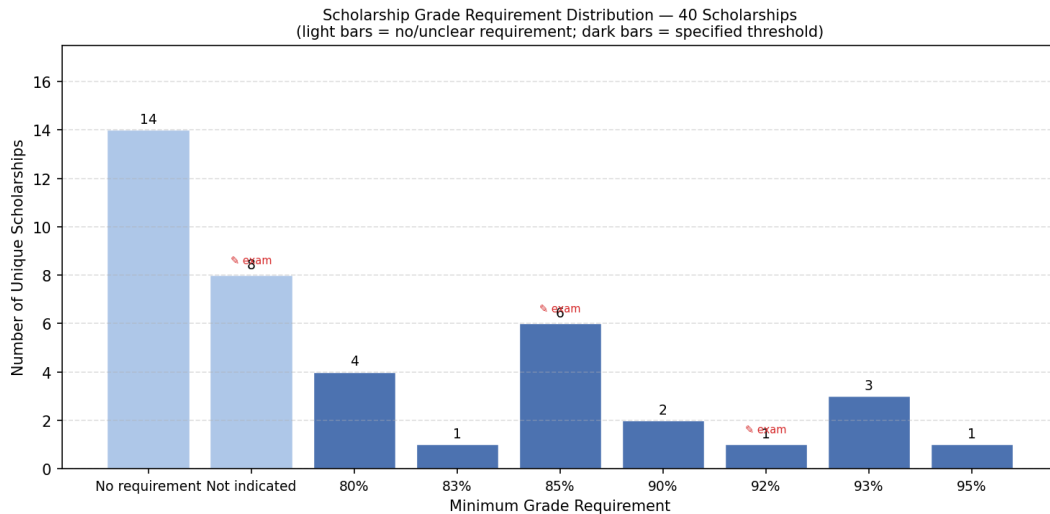


Figure 13: Minimum GWA requirement distribution across 40 scholarships. Light bars indicate no or unclear requirement; dark bars indicate a specified numeric threshold. Bars annotated “exam” require a separate entrance examination (5 of 40 scholarships, 12.5%).

Application period. 17 of 40 scholarships accept applications on a rolling basis (no fixed period). Among those with fixed windows, April is the peak month (7 scholarships), followed by August and June (3 each).

10.5 Recommender integration notes

1. **Q11 filter:** When a student answers Q11 as “nearby only”, the recommender should show scholarship counts only from rows where `nearby_hasSC` = Yes.
2. **Grade filter:** Scholarships with a numeric grade requirement should be filtered against the student’s current GWA (Q-grade item). A student below 85% is ineligible for the majority of scholarships that specify a threshold.
3. **Agriculture & Vet Med priority:** Despite having the sparsest scholarship coverage, Agriculture & Vet Med programs may carry CHED/RDC boosters (Section 8). The recommender should flag the scholarship scarcity and surface any RDC priority boosters to compensate for the perceived lower scholarship accessibility.

10.6 Outputs

- `scholarship_program_bridge.parquet`: cleaned 2,956-row scholarship-program linkage with normalised field names, numeric grade requirement, and binary flags.
- `scholarship_inventory.parquet`: one row per scholarship with type, grade requirement, exam flag, and program counts.
- `reports/eda_v2/tables/tab_scholarship_field_coverage.csv`
- `reports/eda_v2/tables/tab_scholarship_grade_summary.csv`

11 Graduate employment baseline (PIDS 4th GTS)

11.1 Motivation

A program recommender that only ranks fit-to-interest leaves students uninformed about post-graduation employment prospects. Section 8 establishes a national graduate employment prior for each GabayPoz affinity field using the Philippine Institute for Development Studies (PIDS) *4th Graduate Tracer Study* (GTS), Discussion Paper No. 2019-26, covering AY 2009–2011 graduates surveyed in 2014. Although vintage, this remains the most comprehensive nationally representative GTS available for the Philippines and provides the only published in-field employment match rates by discipline.

11.2 Methodology

The GTS reports seven discipline groups: Education, Humanities/Social Sciences/Communication, Business/Management, Natural Science/Agriculture, Engineering/Technology, Health, and Services. These are mapped to the six GabayPoz affinity fields as follows:

- **EDUCATION** ← Education discipline group.
- **ARTS** ← Humanities, Social Sciences & Communication.
- **BUSINESS** ← Business & Management.
- **HEALTH** ← Health Sciences.
- **STEM** ← Weighted average of Natural Science/Agriculture and Engineering/Technology groups, using each group’s share of national graduates as weights (11.5% and 11.1% respectively).
- **AGRICULTURE** ← Proxied from the Natural Science/Agriculture group (best available approximation; the GTS does not disaggregate Agriculture separately).

The *effective employment probability* combines the labour force participation rate (LFPR) and the employment rate within the labour force:

$$p_{\text{employ}} = \text{LFPR} \times \text{Employment Rate}.$$

In-field match rates distinguish PRC-board programmes (observed 49% match rate) from non-PRC programmes (estimated 72% match rate based on GTS degree relevance data). The HEALTH field uses the PRC rate; all other fields use the non-PRC estimate.

11.3 Key findings

- **EDUCATION** has the highest effective employment probability (82.4%) and ranks second in months-to-first-job (6 months), reflecting the consistent public-sector demand for licensed teachers.
- **BUSINESS** ranks first in employment rate (91.6%) with a short time-to-employment (5 months), consistent with broad private-sector demand for accounting and management graduates.
- **HEALTH** has the longest median time to first job (14 months), driven by board examination preparation and the licensure-to-employment pipeline for nursing and allied health programmes.
- **STEM** effective employment probability (78.5%) falls below BUSINESS and EDUCATION despite high demand for engineers, partly because the natural science sub-group (included in the STEM average) has lower employment rates than engineering graduates.

- **AGRICULTURE** effective employment probability (77.9%) is the second-lowest, consistent with a structurally underemployed agricultural labour market. However, the short median time-to-first-job (5 months) suggests that graduates who do enter the labour force find work quickly.
- **ARTS** has the lowest LFPR (78.5%), reflecting higher rates of further study and non-formal employment in social sciences and creative fields.

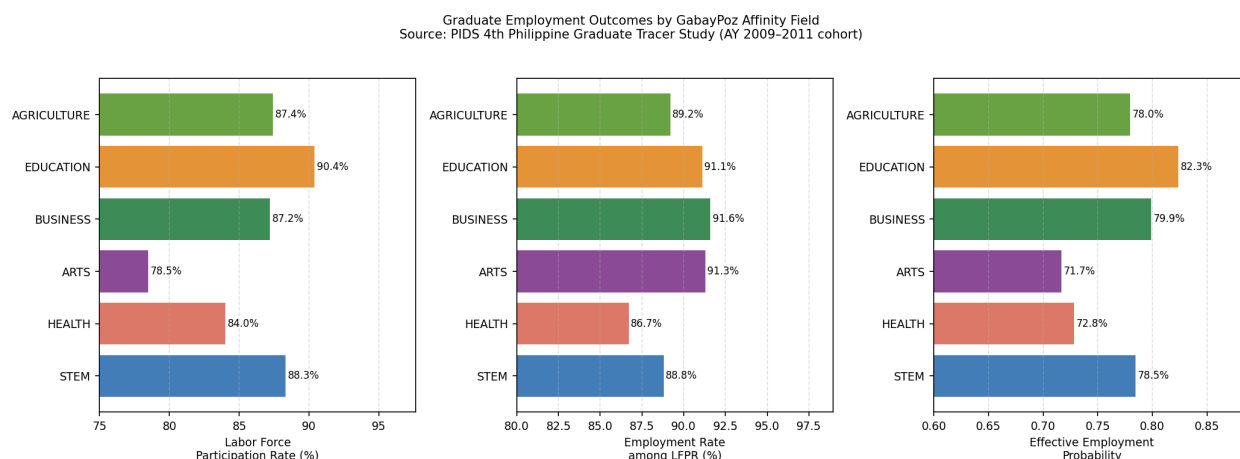


Figure 14: Effective employment probability and in-field match rate by GabayPoz affinity field. Source: PIDS Discussion Paper No. 2019-26 (AY 2009–2011 cohort; surveyed 2014).

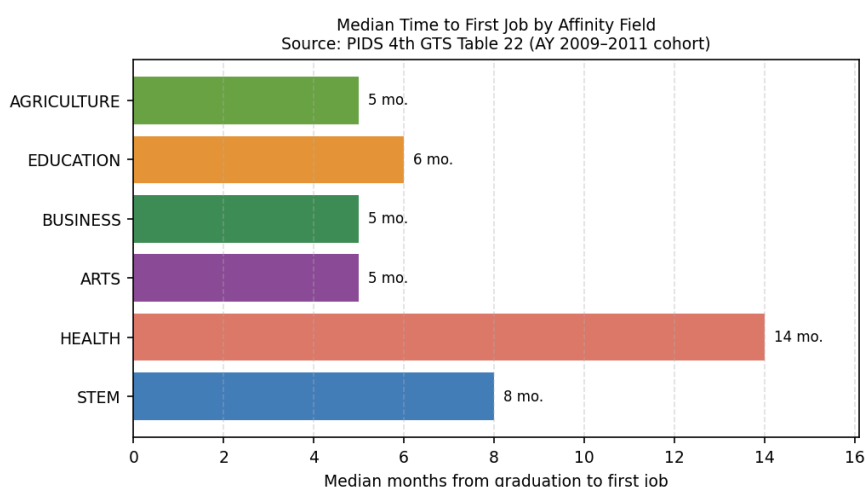


Figure 15: Median months from graduation to first job by affinity field. Source: PIDS 4th GTS.

11.4 Recommender integration notes

The `gts_employment_by_field.parquet` output supplies three explanation fields for the GabayPoz recommendation JSON:

- `effective_employment_probability` — national prior for the programme’s dominant affinity field; used in the explanation block, not in the ranking score.
- `in_field_match_rate` — proportion of graduates working directly in their trained occupation; surfaced in the “career outlook” explanation panel.

- `median_months_to_first_job` — expectation-setting for students; flagged as a caveat for HEALTH programmes (14-month lag).

These values are *national priors*, not local estimates. They should be qualified with the caveat that Pangasinan labour markets may differ and that the GTS vintage (AY 2009–2011) predates major economic shifts (pandemic, industry 4.0 uptake).

11.5 Outputs

- `data/processed/team3_eda/gts_employment_by_field.parquet`
- `reports/eda_v2/figures/fig_gts_employment_rates.png`
- `reports/eda_v2/figures/fig_gts_months_to_first_job.png`

12 OFW viability score (SOF 2024, Region I)

12.1 Motivation

Overseas Filipino Worker (OFW) pathways are a significant component of employment planning for Pangasinan graduates. The GabayPoz database reserves the column `affinity_ofw_viability_score` in the `program` table to capture how strongly each affinity field maps to international labour demand from a Region I context. Section 9 derives this score from the 2024 Survey on Overseas Filipinos (SOF), PSA.

12.2 Methodology

The SOF records each OFW’s 2-digit Philippine Standard Occupational Classification (PSOC) code in column `RQ16_2DGT`. Region I rows are identified by `RREG = 1`. Each 2-digit PSOC sub-major group is mapped to a GabayPoz affinity field following PSOC 2012 group definitions. PSOC groups that span multiple fields (e.g. Managers, Protective Services) are excluded from field-level aggregation.

Weighted OFW counts (`RSWGT`) are summed per field. The share of each field in the total mappable occupation OFW pool is computed, then normalised to the $[0.5, 5.0]$ scale:

$$\text{OFW viability score}_f = \text{clip}\left(\frac{\text{share}_f}{\bar{\text{share}}} \times 2.5, 0.5, 5.0\right),$$

where $\bar{\text{share}} = 1/6$ is the uniform mean share across six fields. A score of 2.5 indicates a field’s share equals the mean; scores above 2.5 indicate above-average OFW representation.

12.3 Recommender integration notes

Once the SOF parquet is generated, `ofw_viability_score` is written into the three reserved columns of the `program` table. In recommender Phase 3, the market dimension weight will shift from the current neutral 1.0 to a blend of local DOLE/PhilJobNet demand (Section 10) and OFW viability, once both are validated.

12.4 Outputs

- `data/processed/team3_eda/ofw_viability_by_field.parquet` (produced once SOF raw data is available)

- `reports/eda.v2/figures/fig_ofw_viability_by_field.png`
- `reports/eda.v2/figures/fig_ofw_occupation_breakdown.png`

Note: the SOF 2024 raw CSV must be extracted from the economics zip archive into `data/raw/` before this section's script can be executed. See `data/raw/README.md` for instructions.

13 Labor demand signal (PhilJobNet 2026)

13.1 Motivation

Graduate tracer data and OFW surveys capture historical and current stock of workers, but not the forward-looking demand signal that employers reveal when they post vacancies. Section 10 operationalises this signal by scraping all current job postings from PhilJobNet (<https://philjobnet.gov.ph>), the official DOLE job vacancy portal, classifying each listing by affinity field, and normalising vacancy shares into demand scores for the two reserved `program` table columns: `affinity_local_demand_score` (Region I listings) and `affinity_national_demand_score` (all listings).

13.2 Data collection and scraping approach

Rather than filtering PhilJobNet by keyword (which returns incomplete subsets), the scraper fetches the unfiltered all-listings page and paginates through all available pages using ASP.NET WebForms `__doPostBack` pagination. For each page, all job cards are parsed from `div.jobcard` elements, extracting: job title (`h1.jobtitle`), employer (`span.companytitle`), location (adjacent to `bi-geo-alt` icon), salary, education requirement, and employment type. The Region I flag is set when the location string contains any of 30 province/city/municipality tokens for Ilocos Norte, Ilocos Sur, La Union, and Pangasinan.

The scrape was conducted on **9 May 2026** and collected **1,960 listings** out of 6,803 reported by the site. The discrepancy likely reflects expired or archived postings counted in the site total but no longer returned in pagination.

13.3 Classification

Job titles are classified into affinity fields using a 6-field keyword dictionary applied in priority order (HEALTH and AGRICULTURE checked first, as their titles are most distinct; BUSINESS last, as it is the broadest). A listing is tagged with the *first* matching field. Of 1,960 listings, 1,209 (61.7%) were classified; 751 (38.3%) were tagged unclassified, primarily representing general-labour roles (e.g. SERVICE CREW, DELIVERY DRIVER, BAGGER) that do not require tertiary education and are thus out-of-scope for a college programme recommender.

13.4 Key findings

- **BUSINESS dominates nationally** (756 of 1,209 classified listings, 62.5%), reflecting the services-oriented structure of the Philippine formal economy and the breadth of business-related job titles. This is consistent with the PIDS GTS employment rate findings (BUSINESS has the highest employment rate and shortest time-to-first-job).
- **STEM ranks second nationally** (209 listings, 17.3%), driven by engineering, technician, and IT support roles.

- **HEALTH ranks third** (154 listings, 12.7%), with nursing and caregiver roles prominent — consistent with sustained post-pandemic demand.
- **ARTS is severely underrepresented** in formal vacancy postings (15 listings, 1.2%), confirming that social science and communications graduates largely enter informal or contract-based employment not captured by PhilJobNet.
- **AGRICULTURE national score is 0.50** (floor), with only 12 listings nationally (1.0%). This starkly contrasts with AGRICULTURE’s actual economic importance — it reflects the near-absence of formal posted vacancies for agriculture-related professionals in an online job board, not an absence of agricultural employment. This finding reinforces the recommendation that AGRICULTURE programme scores should *not* be mechanically penalised using PhilJobNet demand scores in recommender v1.
- **Region I pattern differs from national:** Of 60 Region I listings, 39 were classified. BUSINESS still dominates (29 listings, 74.4%), but AGRICULTURE has a slightly stronger relative presence (2 listings, 5.1% vs. 1.0% nationally). EDUCATION had zero Region I postings in this scrape period, likely due to low vacancy turnover in the public school system (DepEd hires through CSC channels, not PhilJobNet).
- **Region I sample is small** (39 classified listings). The Region I demand scores should be treated as directional indicators only and should not be applied as hard ranking adjustments until the sample grows through repeated quarterly scrapes.

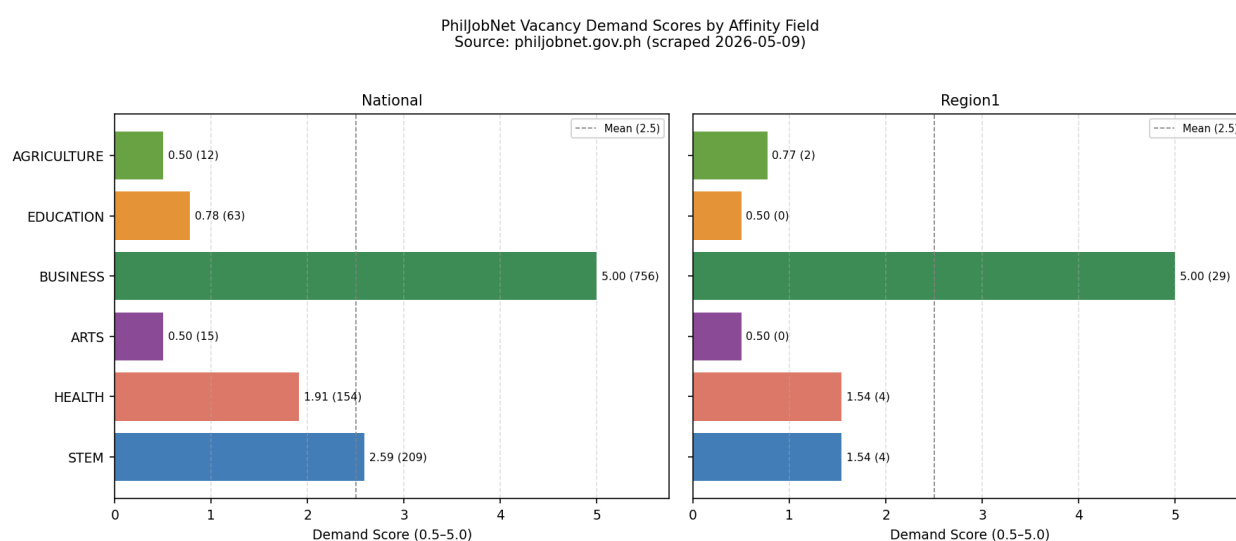


Figure 16: PhilJobNet demand scores by affinity field (national and Region I). Scraped 9 May 2026; $n = 1,960$ listings, 1,209 classified.

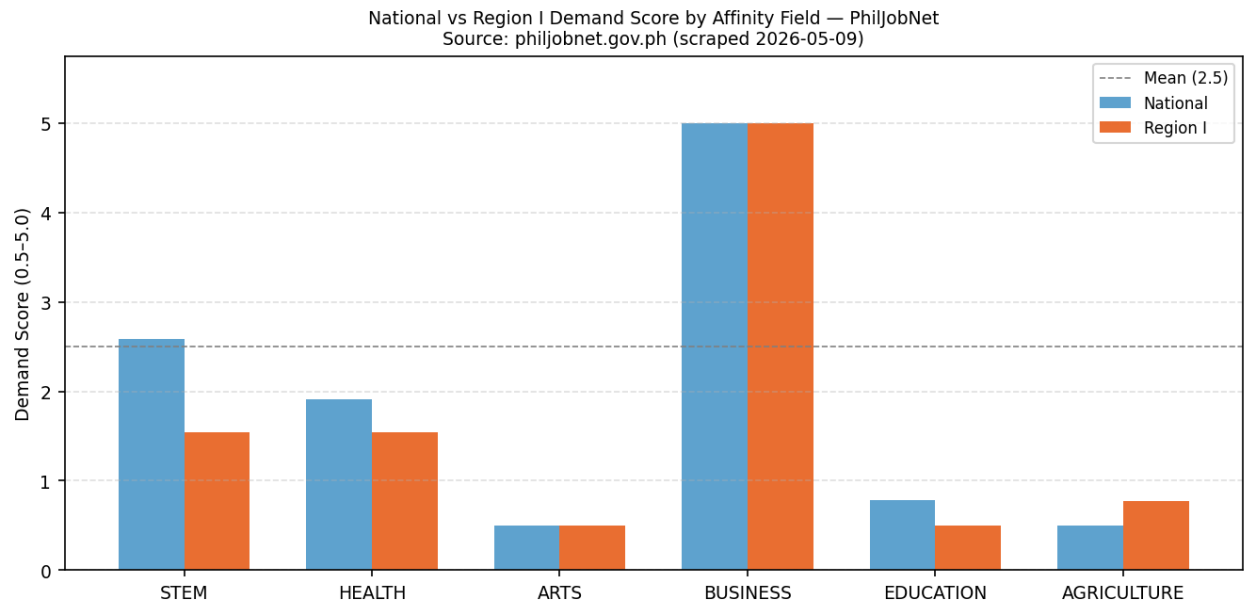


Figure 17: National vs. Region I demand score comparison. AGRICULTURE shows a relatively stronger presence in Region I than in national postings, while EDUCATION has zero Region I postings in the May 2026 scrape.

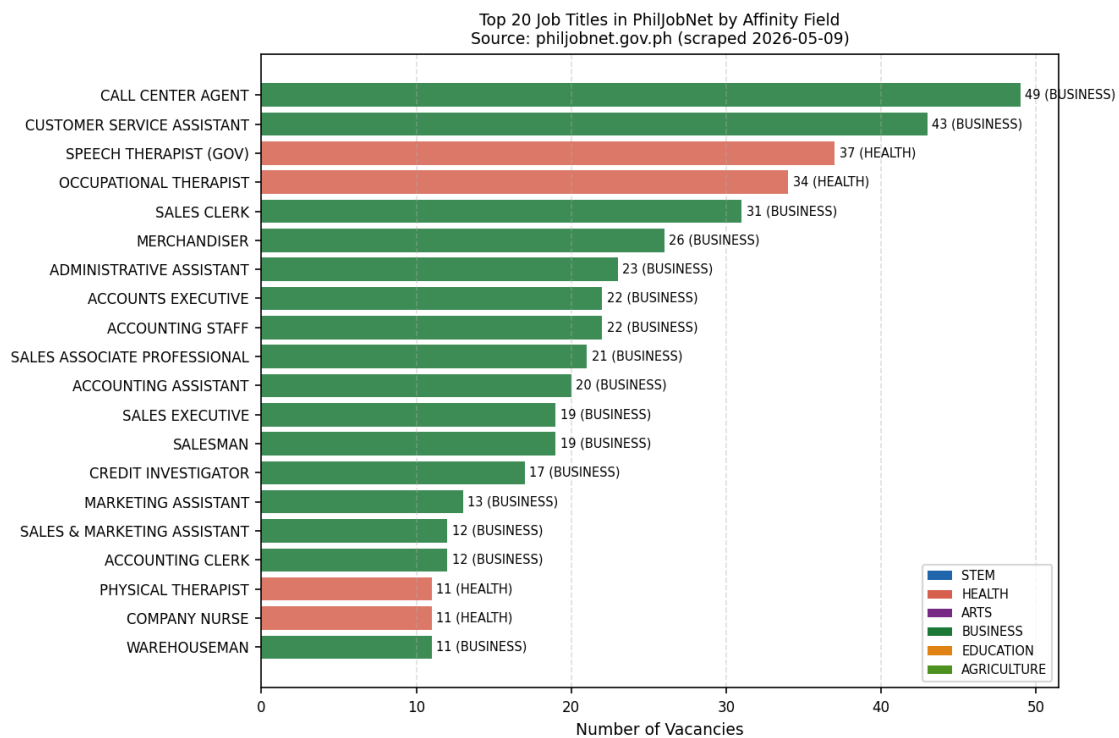


Figure 18: Top 20 classified job titles in the May 2026 PhilJobNet scrape, coloured by affinity field. BUSINESS titles (accountant, HR officer, customer service) dominate; the highest-ranked HEALTH title is company nurse.

13.5 Limitations and recommender integration notes

PhilJobNet captures *formal posted vacancies* only. This systematically undercounts:

- Agricultural employment (informal, seasonal, family-farm basis).
- Teaching positions (DepEd hires via CSC; LGU hires via local board).
- OFW-bound positions (listed on overseas placement agencies, not PhilJobNet).
- Social work and community development roles (NGO and LGU, often referral-based rather than publicly posted).

As a result, the PhilJobNet demand scores **should not be used as the sole signal** for the `affinity_local_demand_score` column. They should be combined with the OFW viability score (Section 9) and the HEAP market signal (Section 3) into a composite demand score once all three are validated. In recommender v1, the market dimension remains at the neutral weight of 1.0.

The scraper should be re-run quarterly. The `--from-cache` flag allows reclassification of cached listings without re-scraping.

13.6 Outputs

- `data/processed/team3_eda/philjobnet_vacancies_raw.parquet`
- `data/processed/team3_eda/philjobnet_demand_by_field.parquet`
- `reports/eda_v2/figures/fig_philjobnet_demand_by_field.png`
- `reports/eda_v2/figures/fig_philjobnet_national_vs_region1.png`
- `reports/eda_v2/figures/fig_philjobnet_top_titles.png`

14 Team 4 handoff summary

Table 4 lists all fifteen parquet artefacts delivered by EDA v2 and their intended use in the GabayPoz recommender.

Table 4: EDA v2 parquet outputs for Team 4 recommender integration.

Parquet file	Contents	Recommender use
barangay_accessibility_summary	Nearest university, counts at 30/60/90 km	Q11 mobility filter
barangay_program_pool_by_threshold	Reachable programs per threshold	Candidate set sizing
q10_tier_calibration	Income quintile PHP breakpoints	Q10 hard filter
barangay_university_economic_burden	Annual tuition + transport burden	Affordability flags
p21_field_bridge	P21 $\rightarrow$ affinity field mapping	Market score computation
heap_occupation_by_field	Pangasinan/Pozorrubio shares; v1 & v2 scores	Market dimension
program_occupation_bridge	CMO-based P21 mapping per program	Explanation fields
program_affinity_profile	6-dimension scores + booster flags	Scoring + boosters
nongainful_by_barangay	Estimated non-gainful counts	Context/explanation
intervention_zones	Below-median-access barangays	Outreach targeting
scholarship_inventory	40 scholarships with attributes	Scholarship filter
scholarship_program_bridge	2,956 scholarship-program rows	Q11 $\times$ scholarship join
<i>New in EDA v2 (May 2026) — employment & demand signals</i>		
gts_employment_by_field	PIDS GTS employment rates, match rates, months-to-first-job per field	Explanation JSON employment outlook block
ofw_viability_by_field	Region I OFW weighted count & viability score per field	affinity_ofw_viability_score column (v2 activation)
philjobnet_demand_by_field	PhilJobNet vacancy counts & demand scores (national + Region I)	Local/national demand score columns (v2 activation)

Schema reference. Full column definitions for each parquet are in `docs/teams/team3_data_engineering_eda.r`

Remaining data gaps. The following items are documented in `docs/reports/team3_eda_data_gaps_notes.v1` and remain open:

- FIES NA semantics for EDUCATION/TRANSPORT/HEALTH spend (audit finding F-3: `fillna(0)` downward bias in weighted means).
- v1.5 audit patches F-1 (fig8 SMALL_N label) and F-2 (tab16 duplicate rows) are documented but not yet applied to the v1 script.
- Per-barangay non-gainful direct measurement requires a Form 3 variant with BGY field, not available in the current CPH 2020 PUF.